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Performance Testing of an Advanced Lightweight Freezable Radiator

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ABSTRACT

During extravehicular activities (EVAs) it is crucial to keep the astronaut comfortable. Currently, a sublimator rejects to space both the astronaut's metabolic heat and that produced by the Portable Life Support System. In doing so, it consumes up to 3.6 kg (8 lbm) of water; the single largest expendable during an eight-hour EVA. While acceptable for low earth orbit, resupply for moon and interplanetary missions will be too costly. Fortunately, the amount of water consumed can be greatly reduced if most of the heat load is radiated to space. However, the radiator must reject heat at the same rate that it is generated to prevent heat stroke or frostbite. Herein, we report on a freezable radiator and heat exchanger to proportionally control the heat rejection rate.

INTRODUCTION

At present, both the astronaut's metabolic heat and that produced by the Portable Life Support System (PLSS) are rejected to space by a sublimator that consumes up to 3.6 kg (8 lbm) of water; the single largest expendable during an eight-hour extravehicular activity (EVA). Unfortunately, this will not be acceptable for moon and interplanetary missions where resupply is difficult. The amount of water lost to sublimation can be greatly reduced by radiating most of the heat load to space. Therefore, TDA Research, Inc. (TDA) developed a lightweight radiator and freezable heat exchanger (RAFT-X) to reject heat. Herein, we report the RAFT-X concept and its performance data.

Figure 1 shows a simplified schematic of an advanced Portable Life Support System (PLSS). The heat of absorption, as well as the heat directly removed from the astronaut's Liquid Cooling Garment (LCG), is carried away by a water loop to the heat rejection system. A sublimator rejects the metabolic heat load and the heat generated by the PLSS to space. Weighing in at only 1.58 kg (3.48 lbm), the sublimator itself is light, but the water tank weighs 2.0 kg (4.4 lbm), and the water weighs an additional 3.6 kg (8 lbm), for a total of 7.2 kg (15.9 lbm [1]). With this system, roughly 2.7 to 3.6 kg (6 to 8 lbm) of water are sublimated (evaporated and lost) during an eight-hour EVA. On extended missions, this is an unacceptably large loss; in fact, it is the single largest expendable in the current extravehicular mobility unit (EMU) [2].

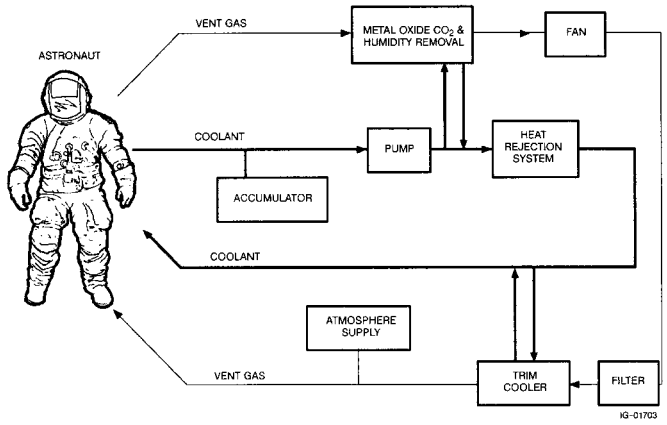


Figure 1.

While the advantages of radiators have long been recognized (they can operate effectively in earth orbit, trans-lunar, trans-Martian, and in most lunar and Mars environments), they have a problem that has so far prevented their use: current simple designs reject heat at a relatively steady rate, but the rate of heat generation by the astronaut is highly variable. Without a way to continuously adjust the heat removal rate, the astronaut will alternate between heat stroke and frostbite conditions.

Figure 2 shows a heat rejection profile for an EVA, which ranges from 100 to 585 W (350 Btu/h to 2000 Btu/h). In addition, thermal loads from the PLSS and heat gains and losses from the environment further extend the range, potentially decreasing the total thermal load by -

30 W (-100 Btu/h) or increasing it by as much as 150 W (500 Btu/h) [3]. Therefore, the total range is from 70 W (250 Btu/h) to 730 W (2500 Btu/h); a turndown ratio of 10:1. Given the relatively small temperature range in which the EMU operates (5 to 24°C (40 to 75°F)), reducing the radiator temperature has little effect on heat rejection. Rather, the effective radiator surface area must be reduced (e.g., by freezing one or more tubes) or thermal resistance increased to match the heat rejection to the astronaut's rate of heat generation.

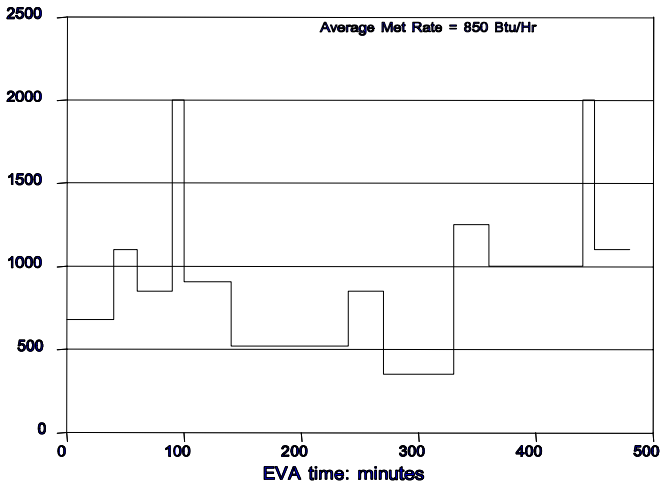


Figure 2. Nominal metabolic profile (Btu/h).

It has long been known that a radiator can be "turned-down" by sequentially allowing tubes that carry the water to the radiator to freeze. When a tube freezes, the temperature of the radiating area around it drops, and the heat rejection rate drops as well. The problem with the freezable radiators developed to date is that they are far too heavy, since they used heavy walled tubes to prevent bursting during freeze/thaw cycles. Further, in past freezable designs, the entire surface of the PLSS had to be thick enough to transfer enough heat to melt the frozen tubes; the radiator must be thick because the heat had to be conducted several inches between adjacent tubes. As a result, past freezable thermal control systems were too heavy.

Cross, Trevino, and Laubach [4] report a system in which the heat flowing to the radiator was decreased by creating a small gap between the radiator and the water loop. When the heat load was high, the gap was filled with a gas to increase the conductance. When the heat load was low, the gap was vented to space, the heat transfer rate was decreased (vacuum is a very good insulator), and the heat rejection rate dropped. However, the weight of the system was 20 kg (44 lbm) compared to 7.2 kg (15.9 lbm) for the current system. Additionally, because of heat leaks, there was only a ~25% reduction in heat rejection in a cold environment.

In order to reduce the system weight, TDA developed the RAFT-X radiator to serve multiple functions. Specifically, it will: 1) reject heat from the PLSS, 2) provide structural resistance to impacts (to cover the components of the PLSS and protect them from damage by micrometeoroids or accidents), and 3) provide a rigid structure to mount equipment. This approach can provide the needed performance with no additional weight penalty. Recently, a flight-weight prototype was built and tested. In this paper, we describe the overall concept for a lightweight freezable radiator and present the experimental performance data obtained in TDA's cold box .

RAFT-X CONCEPT DESCRIPTION

As shown in Figure 3, TDA's approach uses a freeze tolerant heat exchanger in the liquid cooling garment (LCG) water loop to transfer the heat to loop heat pipes that then transfer the heat load to the radiator. Warm water from the astronaut's LCG enters the heat exchanger and transfers heat to the refrigerant loop. The refrigerant boils in the evaporator section of the heat exchanger and vapor moves up the heat pipe that is located on the radiator panel of the PLSS backpack. As heat is rejected by radiation, the vapor condenses and travels back to the evaporator. There are no moving parts (i.e. pumps) and the system is completely passive.

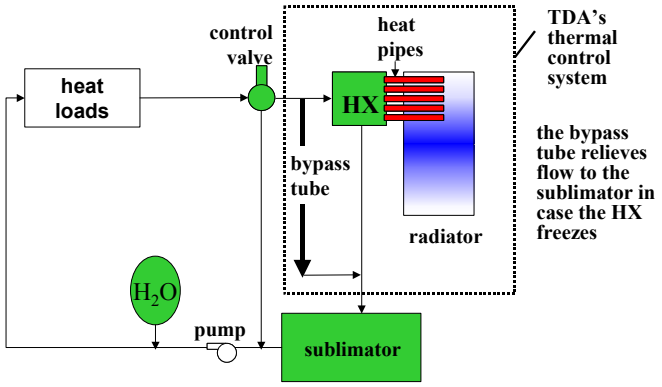


Figure 3. Process flow diagram.

A heat exchanger can accommodate the variable heat load from the crewmember, if it can be selectively frozen to reduce the fin surface area during low load operation. Making it freeze trap tolerant ensures effective and safe operation at any load. Under low metabolic load, the control valve reduces the water flow to the heat exchanger. In cold environments the heat exchanger drops below the freeze point of water and passively controls the heat rejection rate by allowing the coldest water flow passages in the heat exchanger to freeze. As the water in the heat exchanger freezes, sections of the heat exchanger become inactive, preventing heat rejection to the heat pipes that are in contact with those surfaces of the heat exchanger. At the minimum heat load all of the water passages freeze, except for the bypass tube (again, see Figure 3) that will always flow some water to the sublimator and thereby provide continuous and uninterrupted cooling even when the heat exchanger transfers little to no heat. The loop heat

pipes will not freeze even if the PLSS is facing deep space.

There are a number of advantages to this concept. First, using two separate fluid loops allows us to take advantage of the thermal properties of a refrigerant fluid without altering the LCG. The fluid selection has no impact on the rest of the EMU system. Further, because the heat pipe fluid loops are independent from one another and also separate from the water loop, a puncture to one of the heat pipes is not a life-threatening emergency. If the refrigerant in the punctured heat pipe drains out, the water-cooling loop is not affected. The other heat pipes, as well as the sublimator, still remain operational to provide the needed cooling.

A copper heat exchanger (Figure 4) contains both the water loop and the evaporators for the refrigerant system. Its dimensions are 4.17-cm wide x 2.2-cm high x 25.4-cm long and it weighs 1.48 kg (3.26 lbm).

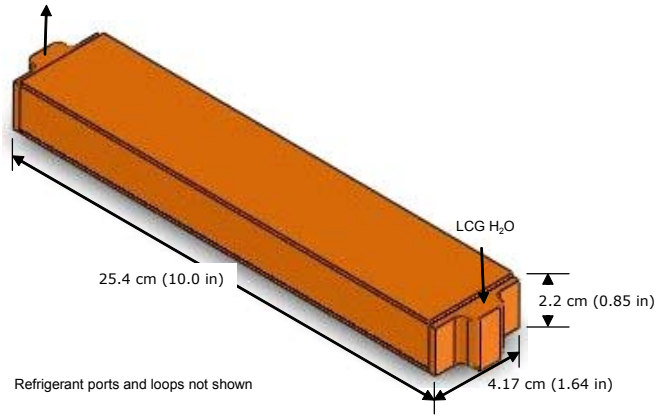
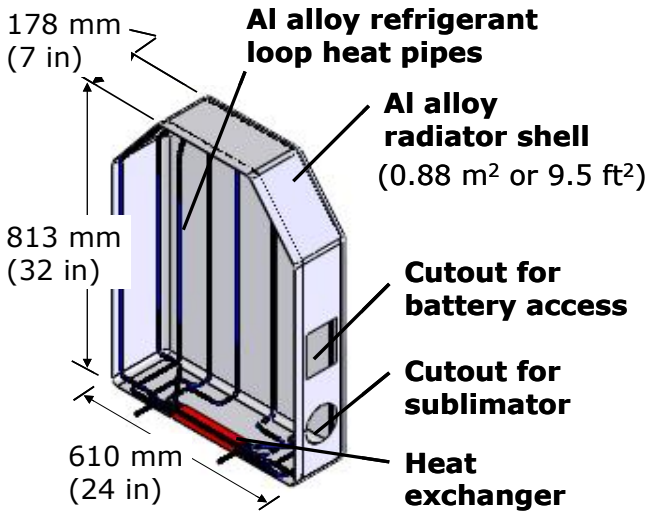


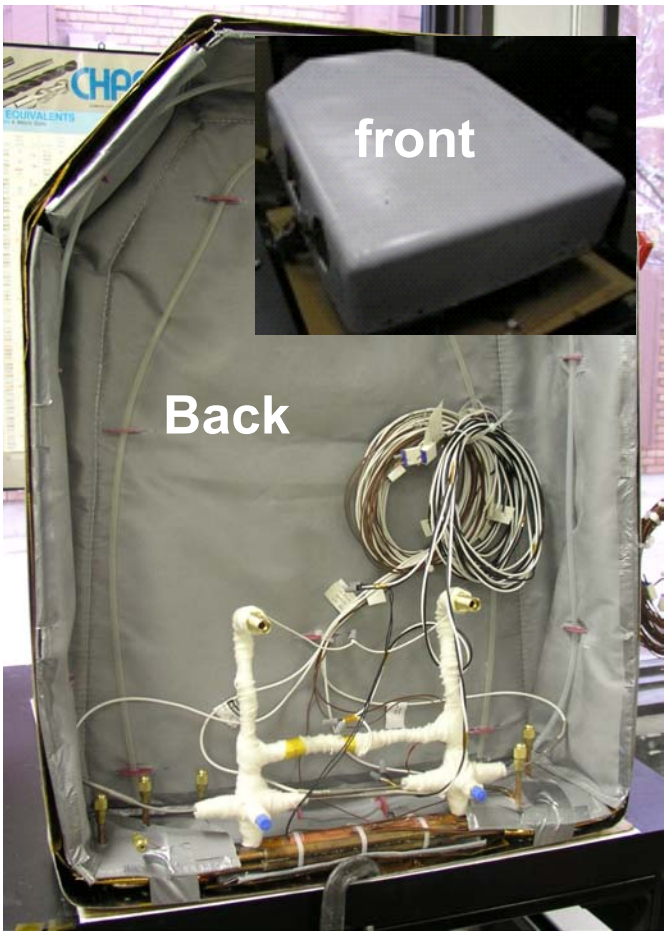
Figure 4. Box size of the freeze tolerant heat exchanger.

The radiator / PLSS shell was constructed from 1.59 mm (0.0625-inch) thick 6061-T6 aluminum sheet and painted with a highly emissive Aeroglaze coating. The Al 6061-T6 radiator wall thickness was sized to provide the needed impact resistance for use on Mars, even though the thermal conductance would have been good enough with much thinner material. TDA installed a 3.05 mm (0.12-inch) thick 7-ply multi-layer insulation for low earth orbit and moon environments. Next, were two 6.35 mm (¼-inch) thick layers of Aspen aerogel SpaceLoft™ AR3103. The aerogel is specifically needed for use in the Mars atmosphere and can also work well in vacuum, but is heavier and thicker than the multilayer insulation. Both were installed to avoid test downtime otherwise needed to repressurize the chamber, swap the MLI and aerogel insulators and once again establish the desired test environment.

The RAFT-X weighs 8.63 kg (19.03 lbm) and Table 1 presents the mass of the major components. The radiator and heat exchanger components are the most massive. In addition, TDA estimated the mass savings from removal of the internal support structure, the EMU thermal protection garment and covers for the secondary oxygen pack, water tank and upper shield. As a result TDA expects a net system savings of 0.17 kg (0.36 lbm) with by implementing the RAFT-X concept.



a) radiator description and dimensions



b) photos of “as built” hardware

Figure 5. RAFT-X prototype radiator.

Table 1. RAFT-X mass breakdown.

Component	Mass, kg (lbm)	C_p, J/kg-C
Radiator w/ emissive paint, loop heat pipes and equipment mounts	5.11 (11.27)	896
MLI and aerogel insulation	1.81 (3.99)	1046
Heat exchanger	1.48 (3.26)	385
Refrigerant	0.23 (0.50)	1050
Total RAFT-X Mass	8.63 (19.03)	
Predicted PLSS mass savings (negative sign to show savings)	-8.80 (-19.39)	
Total Net Mass	-0.17 (-0.36)	

The RAFT-X thermal response to the sink temperature was analyzed assuming that the radiative heat transfer equaled the rate of change in temperature of the mass (Equation 1).

$$mC_p \frac{\Delta T}{\Delta t}\Big|_{RAFT-X} = \varepsilon A \sigma \left(T_R^4 - T_\infty^4\right) \quad (1)$$

Assuming an initial temperature of 21°C and an emissivity of 0.9, it will take about 2 hours to cool the RAFT-X down to a chamber temperature of -40°C (-40°F) and 4 hours to reach -120°C (-185°F) even if the chamber temperature is -156°C (-250°F) as shown in Figure 6. Thus, most test data will be acquired with the RAFT-X still responding to changes in the water flow rate, water inlet temperature and the chamber temperature.

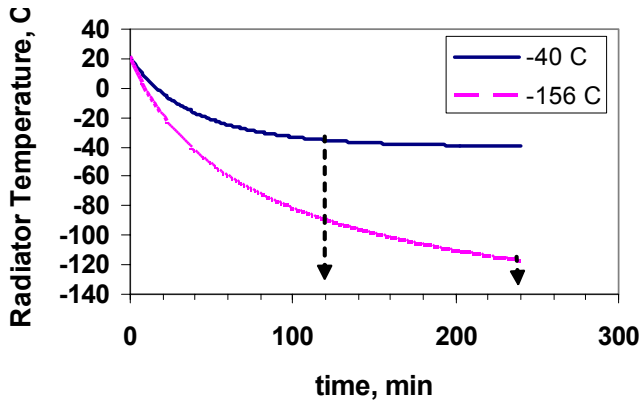


Figure 6. Effect of sink temperature on RAFT-X cooldown.

TEST FACILITIES DESCRIPTION

Twenty-four Type T adhesive thermocouples are attached to the RAFT-X radiator surface as shown in Figure 7. Two dual-element thermistor probes with 44033 elements measure the heat exchanger inlet and outlet water temperatures. Several thermocouples are also attached to the water lines and heat exchanger structure. An absolute pressure transducer monitors the water pressure. A differential strain-gauge pressure sensor measures the pressure drop across the heat exchanger.

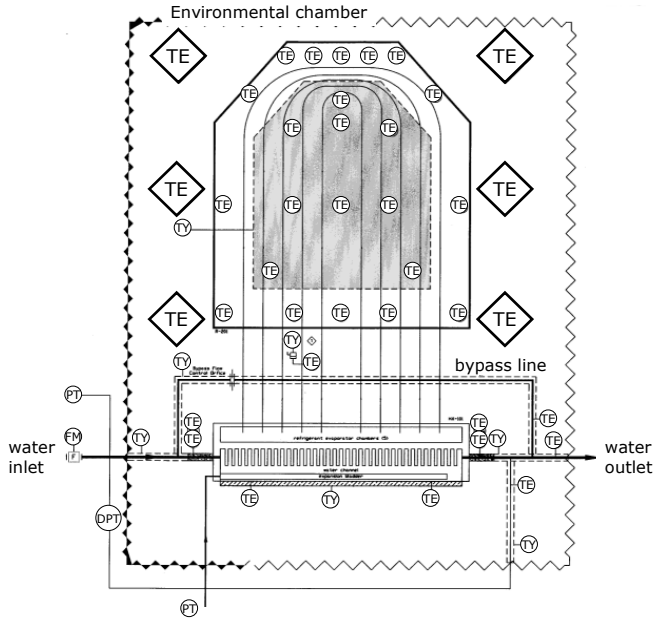


Figure 7. RAFT-X test article instrumentation layout.

Two test facilities were setup to evaluate the RAFT-X performance; a TDA Research cold box for initial operational assessment and the NASA Johnson Space Center (JSC) Environmental Chamber E test facility for critical performance testing in lunar and Mars environments. These are described in greater detail below.

TDA RESEARCH COLD BOX

An approximately 1 m x 1 m x 1 m cold box (Figure 8) was built from 5-cm thick foam to assess RAFT-X operation. The radiator was inclined at 20° to simulate the Mars gravity. A liquid nitrogen cooled aluminum tray above the radiator lowered the sink temperature to as cold as -184°C (-300°F). TDA used Automation Direct Terminator IO hardware with the Entivity Studio Version 7 software to acquire data and control the experiment. All channels were sampled sequentially within a 5 msec window at a 2 Hz data acquisition rate. The thermocouple data were output in engineering units with 16-bit accuracy.



Figure 8. RAFT-X installed in the TDA cold box (front cover removed).

NASA JSC ENVIRONMENTAL CHAMBER E FACILITY

Chamber E shown in Figure 9 has usable dimensions of 1.4 m diameter x 2.9 m (4.6 ft x 9.5 ft) and has liquid nitrogen cooled walls to lower the temperature to -173°C (-280°F). Infrared heat lamps warm the chamber up to 92°C (200°F). The chamber pressure is evacuated down to 10^{-4} Pa (~ 10^{-6} torr) to simulate low earth orbit and lunar conditions. Reduced pressures between vacuum and local atmospheric pressure can also be obtained. For example, JSC can pressurize the chamber to 1333 Pa (10 torr) with argon to simulate the thermal conductance of the Mars atmosphere. Numerous feedthroughs exist for instrumentation, gas and fluid delivery, electric power and mechanical actuation. An in-house Data Acquisition and Recording System (DARS) records and stores data at 1 Hz with 16-bit precision. All channels are sampled sequentially at a 50 kHz clock speed. LabView operates through National Instruments boards to control the facility and test article.



Figure 9. NASA JSC Chamber E. [5]

NASA JSC plans to subject the RAFT-X to various sink temperatures and heat loads in order to simulate astronaut metabolic loading in lunar and Mars EVA environments. The objective of each test point will be to determine the RAFT-X heat rejection capability at steady state conditions.

De-ionized water will be continuously pumped through the heat exchanger at flow rates between 4.5 and 90 kg/h (10 and 200 lbm/h) and at a pressure of 207 kPa (30 psia). A Micro Motion™ coriolis flow meter will measure the water flowrate. All of the water supply lines within the chamber will be heat traced and covered with vacuum compatible radiative insulation in order to prevent freezing. Power to the heat tracing will be remotely controlled and monitored.

An infrared lamp array comprising seventeen 1000-watt lamps (Figure 10) will adjust the environmental temperature to within ±6°C (±10°F) over a wide range of temperatures from -156 to +38°C (-250 to +100°F). Type T thermocouples attached to six suspended coupons will provide a continuous survey of the Chamber E environment during the tests.

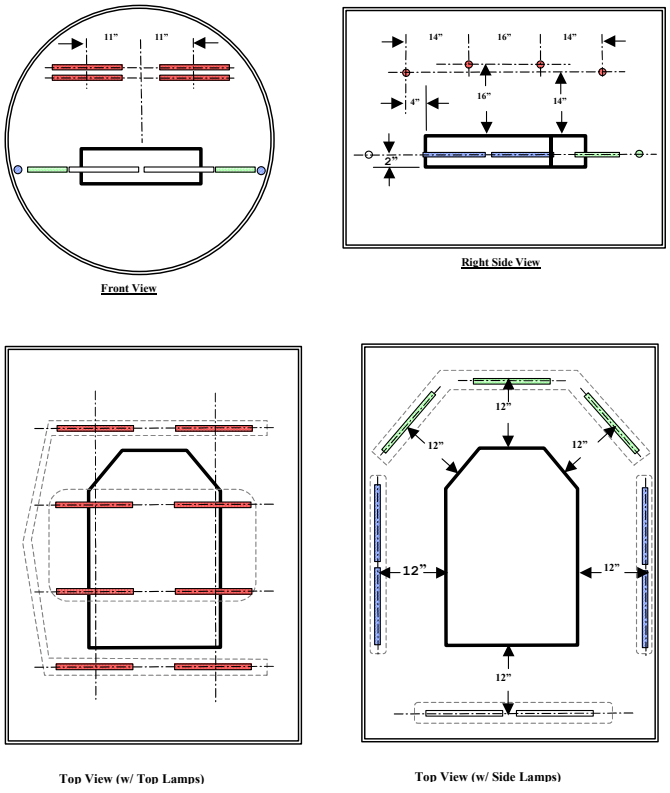
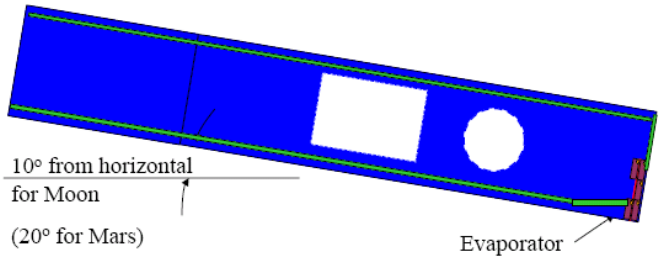


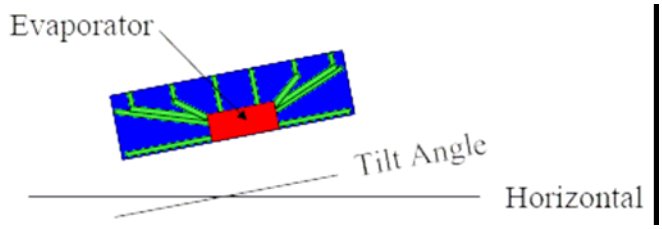
Figure 10. IR lamp placement to establish the desired environmental temperature.

The NASA JSC Systems Test Branch built the RAFT-X test article test stand. This structural support is well insulated to provide thermal isolation from the test article, and was designed to not interfere with the environmental conditioning. In addition, the support structure will provide attitude control in two planes (both

inclination and lateral tilt as sketched in Figure 11). The angles of inclination to simulate lunar and Mars gravities are 10° and 20° at earth gravity, respectively. Similarly, the lateral tilt angles of 2.5° and 5° correspond to the maximum 15° angle that an astronaut may lean without falling over in lunar and Mars gravities, respectively. These will be simulated to demonstrate radiator operation independent of the radiator and heat exchanger position relative to the gravity vector.



a) inclination angles to simulate lunar and Mars gravities



b) lateral tilt to simulate astronaut lean

Figure 11. Test article positioning relative to earth's gravity vector to simulate lunar and Mars gravities.

Several tests are planned to fully evaluate the RAFT-X operability in lunar and Mars environments. We will determine the maximum and nominal heat rejection in those environments, conduct life cycle tests while varying the heat load from 60 to 300 W (~ 200 to 1000 Btu/h) and demonstrate freeze tolerance (including the freeze trap scenario). Performance degradation due to the loss of a heat pipe will also be tested.

TEST RESULTS AND DISCUSSION

Primary objectives of this test program were to demonstrate RAFT-X operation and measure performance in simulated lunar and Mars thermal environments. Secondary objectives were to demonstrate heat exchanger freeze-tolerance and radiator restart after a freeze-up.

TDA OPERATIONAL AND PERFORMANCE ASSESSMENT TESTS

Initially, TDA conducted leak checks of the heat exchanger and the loop heat pipes. TDA measured the pressure drop of the heat exchanger waterside flow at 90 kg/h (200 lbm/h) to ensure that it was less than 14 kPa (2 psi). This is the nominal pressure increase from the LCG water pump. Then, TDA conducted operability tests in the cold box (again, see Figure 8) to ensure that the prototype RAFT-X functioned properly before shipment to NASA. The water flow remained laminar through the heat exchanger even though the flow rate was varied from 22.7 to 90.7 kg/h (50 to 200 lbm/h). The local atmospheric pressure was 84.1 kPa (12.2 psia). Sink temperatures from ambient to about -156°C (-250°F) were established by spraying LN_2 over the aluminum cold plate.

Up to 225 W (770 Btu/h) were rejected by the RAFT-X at the flow conditions tested. Heat transfer increased with water flow rate (Figure 12). Throttling the water flow reduced the heat transfer to the radiator. Because the cold box was not evacuated, convection and conduction mechanisms also transferred heat from the heat exchanger and radiator surfaces to the surroundings. Thus, the heat rejected by the radiator was usually only about ½ of the total heat transferred from the heat exchanger.

The heat transfer rate was lowered by reducing the water flow rate as already mentioned. This also initiated freeze up of the water flow channel to further reduce the rate of heat transfer. The increased thermal resistance by the buildup of ice as the water flow channel froze also reduced the rate of heat transfer.

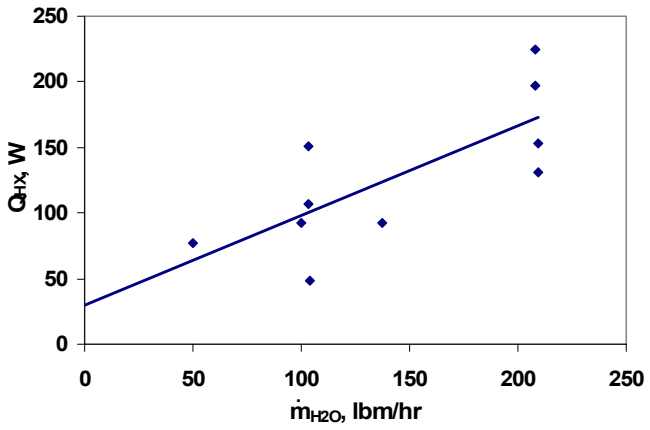


Figure 12. Controlling water flow rate to vary heat rejection.

The heat exchanger water flow channel was frozen solid without rupture and with little increase in system pressure, since the water bypass remained open. The pressure drop across the heat exchanger was no more

than 690 Pa (0.1 psid) when water flowed freely, but the pressure drop increased to about 27,600 Pa (4 psid) when the water channel froze and all of the flow passed through the bypass tube (Figure 13). Test data with a ΔP in between these two limits is further evidence that the heat exchanger operated in a partially frozen state to proportionately lower the heat transfer rate.

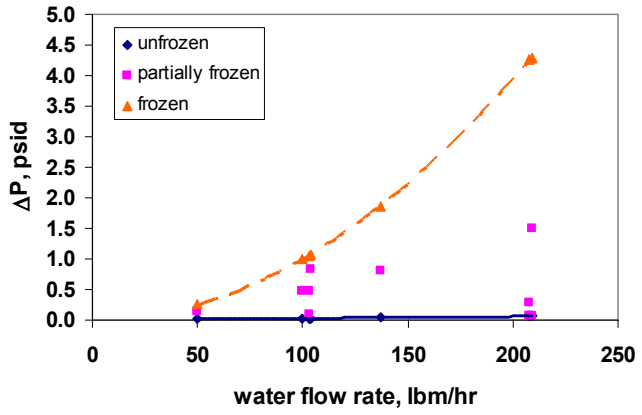


Figure 13. Pressure loss across the heat exchanger when flowing freely, partially frozen and completely frozen (all water flows through the bypass tube).

The heat rejection fell to about 50W once the water flow channel was frozen (Figure 14). The flow through the heat exchanger when partially frozen was deduced by comparing the measured pressure drop across the heat exchanger to the water flow calibration data obtained in both the fully frozen (i.e. bypass mode) and completely unfrozen states. Using the pressure drop measurement, TDA predicted the water flow through the bypass tube, which was then entered into Equation 2 to calculate the quality or percent frozen with about 4% uncertainty. About 80% of the water flows through the heat exchanger and 20% through the bypass tube when completely thawed.

$$quality = \left(\frac{\dot{m}_{bypass}}{\dot{m}_{H_2O}} - 0.2 \right) \cdot \frac{100\%}{0.8} \qquad (2)$$

A flow coefficient, C_v, was empirically determined from Equation 3. TDA assumed that the pressure drop remained proportional to the dynamic pressure. The flow coefficient was estimated to be about 0.7.

$$C_v = \frac{\dot{m}}{\rho A \sqrt{\frac{2\Delta P}{\rho}}} \qquad (3)$$

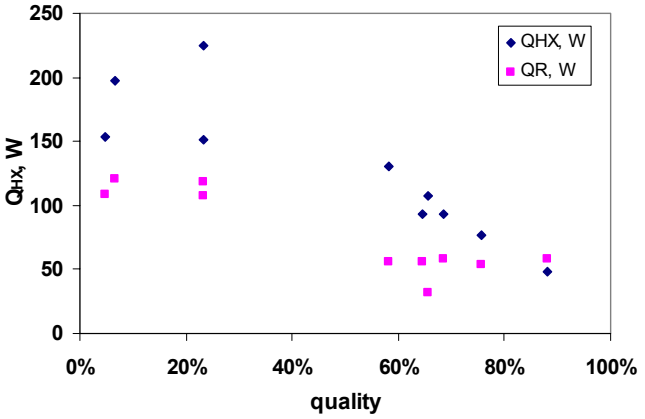
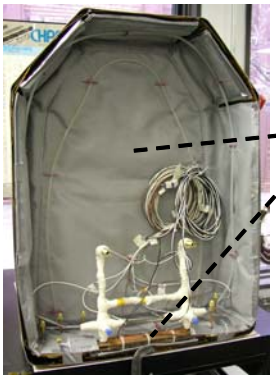


Figure 14. Throttling the heat rejection by freezing up the water side of the heat exchanger.

In Figure 15 TDA presents the average temperature data for two heat rejection loads to a cold environment. The water flow channel in the heat exchanger was nearly frozen for a low power test condition in which the heat exchanger was -45°C (-49°F). The inlet thermistor was also below freezing and was probably coated in ice. The heat exchanger also dropped below the freeze point of water at the higher heat load, so the water flow channel was partially frozen to match the water convective heat transfer rate to the radiation and convection from the radiator surface. For both test conditions the outlet water temperature was only a couple of degrees colder than the inlet.



	50W	225W
ṁ, kg/h	47	94
% frozen	88	23
T_∞, °C	-120	-134
T_R, °C	-67	-44
T_{HX}, °C	-45	-9
$T_{H2Oinlet}$, °C	-2	+11
ΔT_{water}, °C	2.5	2.3

Figure 15. Average temperature data for low and high heat rejection loads.

NASA JSC CHAMBER E TEST PLANS

Unfortunately, a recent stop work order at NASA JSC has delayed the test entry into Chamber E. As outlined in Table 2, several tests were planned to measure the RAFT-X performance in lunar and Mars environments with rejection temperatures from +38 down to -156°C (+100 to –250°F). A hot environment test will ensure that the radiator does not transfer heat from the environment to the astronaut. Thermal cycle tests are

expected to show ability of the RAFT-X to adjust its heat rejection to the input load, which will be varied from 60 to 300 W. The performance degradation due to the loss of a heat pipe will also be determined. For this test, the refrigerant will be remotely vented from a single loop heat pipe to simulate a leak due to micro meteoroid impact. NASA JSC will also simulate the freeze trap scenario in the heat exchanger due to water pump failure. Afterwards, test personnel will thaw the heat exchanger and repeat a maximum heat rejection test to ensure full recovery without loss of performance.

Table 2. RAFT-X Preliminary Test Points.

TEST TYPE	TEST PT. DESCRIPTION	CHAMBER ENVIRONMENT		T_{H2O} (°C)	WATER FLOWRATE (kg/h)
		DESCRIPTION	T (°C)/P (Pa)		
Hot Environment	G1. Heat Gain Test	Ambient	+38/Ambient	18	90.7
Performance Tests, Lunar (Steady State)	L1. Max Heat Rej, No Tilt L2. Max Heat Rej, Lat Tilt	Cold Lunar Cold Lunar	-156/HV -156/HV	18 18	90.7 90.7
Life Cycle Tests (Low heat lead for 60 min., high heat load for 30 min.)	L4. Freeze/Thaw, cycle 1 L5. Freeze/Thaw, cycle 2 L6. Cycle 3 if needed L3. Avg Heat Rej, No Tilt	Cold Lunar Cold Lunar Cold Lunar Nominal Lunar	-156/HV -156/HV -156/HV -34/HV	24/18 24/18 24/18 18	4.5/90.7 4.5/90.7 4.5/90.7 90.7
Performance Tests, Mars (Steady State)	M1. Max Heat Rejection M2. Max Heat Rej, Lat Tilt M3. Avg Heat Rejection	Cold Mars Cold Mars Nominal Mars	-100/1333 -100/1333 -45/1333	18 18 18	90.7 90.7 90.7
Repeat Additional Tests as Needed					
Loss of Single Heat Pipe Steady State Test, Mars	M4. Degraded Max Heat Load	Cold Mars	-100/1333	18	90.7
Loss of Single Heat Pipe Steady State Test, Lunar	L7. Degraded Max Heat Load	Cold Lunar/4 hrs	-156/HV	18	90.7
Pump Failure Simulation	L8. Freeze-up Test During 90 min. EVA emergency	Cold Lunar/TBD	-156/HV	N/A	0
Wrap-up, Repeat as Needed					

CONCLUSION

In conclusion, a lightweight and freeze tolerant heat exchanger can reject about half of the heat generated by the astronaut and EMU during an EVA. The heat rejection was largely limited by the available surface area of the EMU. In performance tests at TDA, the RAFT-X demonstrated total heat rejection from 50 to 225 W; of which about half was attributed to radiation. The RAFT-X functioned through numerous heat rejection cycles from low to high loads. TDA throttled the heat rejection by varying the flow rate and freezing up the water flow channel. Also, TDA was able to thaw and then restart the heat exchanger after a hard freeze. These test results demonstrate the robust heat rejection performance that the RAFT-X concept can provide to an astronaut during EVA.

Although final thermal-vacuum tests were not conducted, a test entry into Chamber E has just been scheduled for the first week of June. The facility will have much needed control of the environmental test parameters to better quantify RAFT-X performance.

ACKNOWLEDGMENTS

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DEFINITIONS, ACRONYMS AND ABBREVIATIONS

ACRONYMS

EMU: Extravehicular Mobility Unit

EVA: Extravehicular Activity

IO: Input Output

LCG: Liquid Cooling Garment

PLSS: Portable Life Support System

RAFT-X: Radiator and Freeze Tolerant Heat Exchanger

NOMENCLATURE

A	surface area or cross-sectional flow area, m^2 (ft^2)
C_p	specific heat, J/kg-K (Btu/lbm-R)
C_v	flow coefficient
DPT	differential pressure transducer
FM	flow meter
HX	heat exchanger
LN_2	liquid nitrogen
m	mass, kg (lbm)
$\dot{m}$	mass flow rate, kg/h (lbm/h)
ΔP	pressure drop, Pa (psid)
PT	pressure transducer
Q	heat rejection rate or power, W (Btu/h)
Δt	time difference
ΔT	temperature difference
T	temperature, °C (°F)
TE	temperature element, either thermocouple or thermistor
TY	temperature control
ε	emissivity
ρ	density, kg/m^3 (lbm/ft^3)
σ	Stefan-Boltzmann constant, 5.67E-08 W/m^2-K^4

Subscripts

bypass	the heat exchanger bypass tube
R	radiator
∞	ambient surroundings, cold plate or sink